

- Take a screenshot showing the completion of the command including its output

```

}

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + ec2instance = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.guestbook: Creating...
aws_instance.guestbook: Still creating... [10s elapsed]
aws_instance.guestbook: Creation complete after 13s [id=i-0b1af33f6c9669f21]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:
ec2instance = "44.201.81.179"

```

- Take a screenshot that includes the VM's IP addresses

> i-0b1af33f6c9669f21

<

Instance summary for i-0b1af33f6c9669f21 Info

Updated less than a minute ago

Instance ID i-0b1af33f6c9669f21	Public IPv4 address 44.201.81.179 open address	Private IPv4 addresses 172.31.81.182
IPv6 address -	Instance state Running	Public IPv4 DNS ec2-44-201-81-179.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-81-182.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-81-182.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t2.micro	AWS Compute Optimizer finding Optimize AWS Compute Optimizer for...
Auto-assigned IP address 44.201.81.179 (Public IP)	VPC ID vpc-0e37e1c50c434503 (Public)	

- Take a screenshot of the successful ssh login from Cloud Shell.

```
[cloudshell-user@ip-10-134-52-121 tf]$ ssh ubuntu@3.84.253.171
The authenticity of host '3.84.253.171 (3.84.253.171)' can't be established.
ED25519 key fingerprint is SHA256:SwO5tPhYjENyWMOLyHgjaO1vWPIiShA0sFrLQ0Ui+pA.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.84.253.171' (ED25519) to the list of known hosts.
Enter passphrase for key '/home/cloudshell-user/.ssh/id_ed25519':
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 5.11.0-1022-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Thu Feb 20 21:44:36 UTC 2025

System load:  0.01          Processes:            106
Usage of /:   18.3% of 7.69GB Users logged in:      0
Memory usage: 21%          IPv4 address for eth0: 172.31.85.132
Swap usage:   0%

1 update can be applied immediately.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
```

- Take a screenshot of the output of the command that includes the IP address of the instance

```
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_key_pair.kp: Creating...
aws_security_group.sg-guestbook: Creating...
aws_key_pair.kp: Creation complete after 0s [id=guestbook-key]
aws_security_group.sg-guestbook: Creation complete after 3s [id=sg-00da86677db44c755]
aws_instance.guestbook: Creating...
aws_instance.guestbook: Still creating... [10s elapsed]
aws_instance.guestbook: Creation complete after 12s [id=i-09b692bd8af069869]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

ec2instance = "3.92.77.28"
```

- Take a screenshot of the Guestbook including the URL with the entry in it.

Guestbook

Sign [here](#)

Entries

Joseph Gafron <joeygafron@gmail.com>
signed on 2025-02-20
Hello Terraform on AWS!

- Take a screenshot that includes the VM's IP addresses

[←](#)
[tf-lab-vm](#)
[EDIT](#)
[RESET](#)
[+ CREATE MACHINE IMAGE](#)
[⋮](#)
[EQUIVALENT CODE](#)
[LEARN](#)

DETAILS

OBSERVABILITY

OS INFO

SCREENSHOT

None

Network interfaces

Name ↑	Network	Subnetwork	Primary internal IP address	Alias IP ranges	IP stack type	External IP address
nic0	default	default	10.138.0.22		IPv4	None

Storage

- Take a screenshot showing the completion of the command including its output

```

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

google_compute_address.static: Creating...
google_compute_address.static: Still creating... [10s elapsed]
google_compute_address.static: Creation complete after 11s [id=projects/cloud-gafron-jgafron/regions/us-west1/addresses/ipv4-address]
google_compute_instance.default: Modifying... [id=projects/cloud-gafron-jgafron/zones/us-west1-b/instances/tf-lab-vm]
google_compute_instance.default: Still modifying... [id=projects/cloud-gafron-jgafron/zones/us-west1-b/instances/tf-lab-vm, 10s elapsed]
google_compute_instance.default: Modifications complete after 10s [id=projects/cloud-gafron-jgafron/zones/us-west1-b/instances/tf-lab-vm]

Apply complete! Resources: 1 added, 1 changed, 0 destroyed.

Outputs:
ip = "34.127.74.3"

```

- Take a screenshot that includes the VM's IP addresses

☐	☒	tf-lab-vm	us-west1-b	10.138.0.22 (nic0)	34.127.74.3 (nic0)	SSH	⌵	⋮
☐	☐	vm europe-west1-d	europe-west1-d	10.138.0.2		SSH		⋮

- Take a screenshot of the successful ssh login from Cloud Shell.

```

jgafron@cloudshell:~/tf/tf (cloud-gafron-jgafron)$ ssh 34.127.74.3
Enter passphrase for key '/home/jgafron/.ssh/id_ed25519':
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1075-gcp x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Feb 20 22:37:55 UTC 2025

System load:  0.08               Processes:            102
Usage of /:   20.2% of 9.51GB    Users logged in:     0
Memory usage: 5%                IPv4 address for ens4: 10.138.0.22
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

```

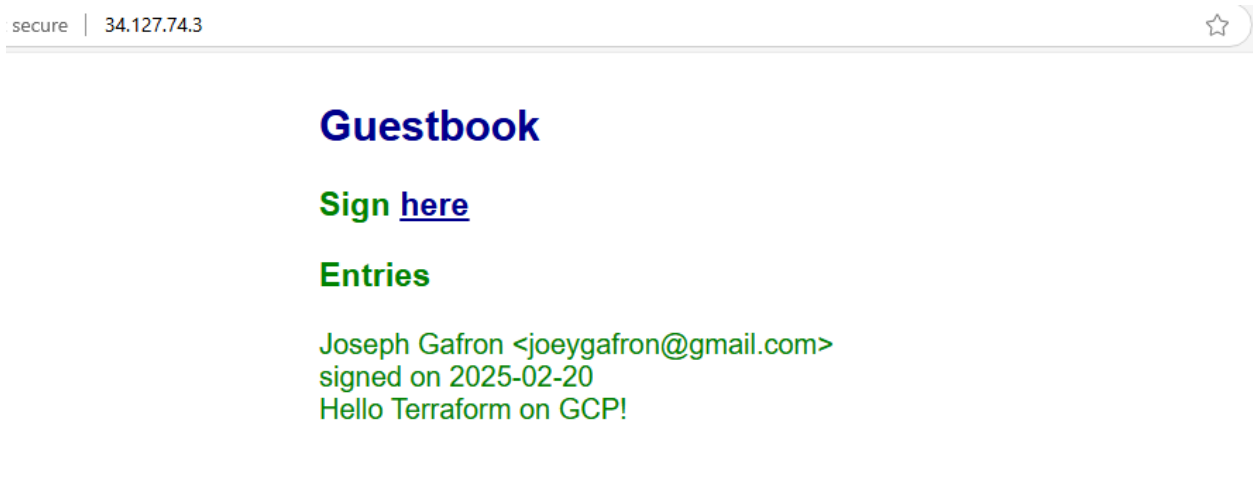
- What resources are being added, changed, or destroyed?

We added a network tag, which allows http traffic through a firewall rule. We modified the metadata-startup-script that runs install.sh on the instance, along with some dependencies and the flask app.

- **What part of the configuration forces a replacement to occur?**

The metadata-startup-script portion of the configuration file forces the replacement.

- **Take a screenshot of the Guestbook including the URL with the entry in it.**



- **What is the name of the Instance Template dynamically generated to create the two nodes (VMs)?**

[gke-guestbook-default-pool-8b4f74bc](#)

- **What is the name of the Instance Group dynamically generated that the two nodes belong to?**

[gke-guestbook-default-pool-8b4f74bc-grp](#)

- **What are the names of the two nodes?**

[Gke-guestbook-default-pool-8b4f74bc-fth2](#)

[Gke-guestbook-default-pool-8b4f74bc-2f97](#)

- Take a screenshot of the container image created

← 7c0bccd56fbed DELETE SETUP INSTRUCTIONS DEPLOY REFRESH SHOW ALL VERS

gcr.io > cloud-gafron-jgafron > gcp_gb > sha256:7c0bccd56fbedf3f2a101acb1465c89ccd889e888e1f90e779946f8850e76555

OVERVIEW **PULL** **MANIFEST** **FILES** **ATTACHMENTS**

Format Docker

Media type application/vnd.docker.distribution.manifest.v2+json

Project cloud-gafron-jgafron

Location us (multiple regions in United States)

Repository gcr.io

Image gcp_gb

Digest sha256:7c0bccd56fbedf3f2a101acb1465c89ccd889e888e1f90e779946f8850e76555

Virtual size 1.2 GB

Built Feb 20, 2025, 4:56:49 PM

Created Feb 20, 2025, 5:01:57 PM

Updated Feb 20, 2025, 5:01:57 PM

Tags latest

Subject digest —

Artifact type —

- Take a screenshot of the output of the following command when all 3 replicas reach a "Running" state.

```
jgafron@cloudshell:~/cs430-src/05_gcp_datastore (cloud-gafron-jgafron)$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
guestbook-replicas-pdz22            1/1     Running   0           2m55s
guestbook-replicas-t4kz9            1/1     Running   0           2m55s
guestbook-replicas-ww5z1            1/1     Running   0           2m55s
```

- Take a screenshot of listing services with LoadBalancer indicating an external IP address that is ready for access.

```
jgafron@cloudshell:~/cs430-src/05_gcp_datastore (cloud-gafron-jgafron)$ kubectl get services
NAME            TYPE           CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
guestbook-lb    LoadBalancer  34.118.225.77 35.233.216.234 80:31943/TCP     3m23s
kubernetes      ClusterIP      34.118.224.1  <none>        443/TCP          23m
```

- Take a screenshot of the Guestbook including the URL with the entry in it.

[Sign here](#)

Entries

Joseph Gafron <joeygafron@gmail.com>
signed on 2025-02-06 07:19:17.908081+00:00
Hello Compute Engine!

Joseph Gafron <joeygafron@gmail.com>
signed on 2025-02-06 06:53:06.106613+00:00
Hello Cloud Shell!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 06:41:06.371140+00:00
Hello Datastore!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-12 01:48:46.343697+00:00
Hello App Engine!

Joseph Gafron <joeygafron@gmail.com>
signed on 2025-02-06 07:19:02.526088+00:00
Hello Cloud Shell!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 06:42:58.746814+00:00
Hello Docker Datastore!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-12 07:36:13.859830+00:00
Hello Cloud Run!

Joseph Gafron <joeygafron@gmail.com>
signed on 2025-02-21 01:10:03.185674+00:00
Hello Kubernetes!

-
- Take a screenshot of the managed guestbook pods and the service being exposed.

Managed pods

Name	Status	Restarts	Created on ↑
guestbook-replicas-ww5zl	✓ Running	0	Feb 20, 2025, 5:05:01 PM
guestbook-replicas-t4kz9	✓ Running	0	Feb 20, 2025, 5:05:01 PM
guestbook-replicas-pdz22	✓ Running	0	Feb 20, 2025, 5:05:01 PM

Exposing services ?

Name ↑	Type	Endpoints
guestbook-lb	Load balancer	35.233.216.234:80 ↗

- Take a screenshot of the load balancer and its details

a40cff9d961224d4896f631e5500674d

Target-pool Network Load Balancer

Frontend

Protocol ↑	IP version	IP:Port	Network Tier ?
TCP	IPv4	35.233.216.234:80	Premium

Backend

Name	Region	Health check
a40cff9d961224d4896f631e5500674d	us-west1	k8s-ee07658b555af771-node

✓ ADVANCED CONFIGURATIONS

Instance ↑	Zone	35.233.216.234
gke-guestbook-default-pool-8b4f74bc-2f97	us-west1-b	✓
gke-guestbook-default-pool-8b4f74bc-fth2	us-west1-b	✓

- Take a screenshot of the addresses allocated and indicate the ones associated with nodes versus the one associated with the load balancer.

These are the ones associated with nodes:

VM instance [gke-guestbook-default-pool-8b4f74bc-2f97](#) (Zone us-west1-b)

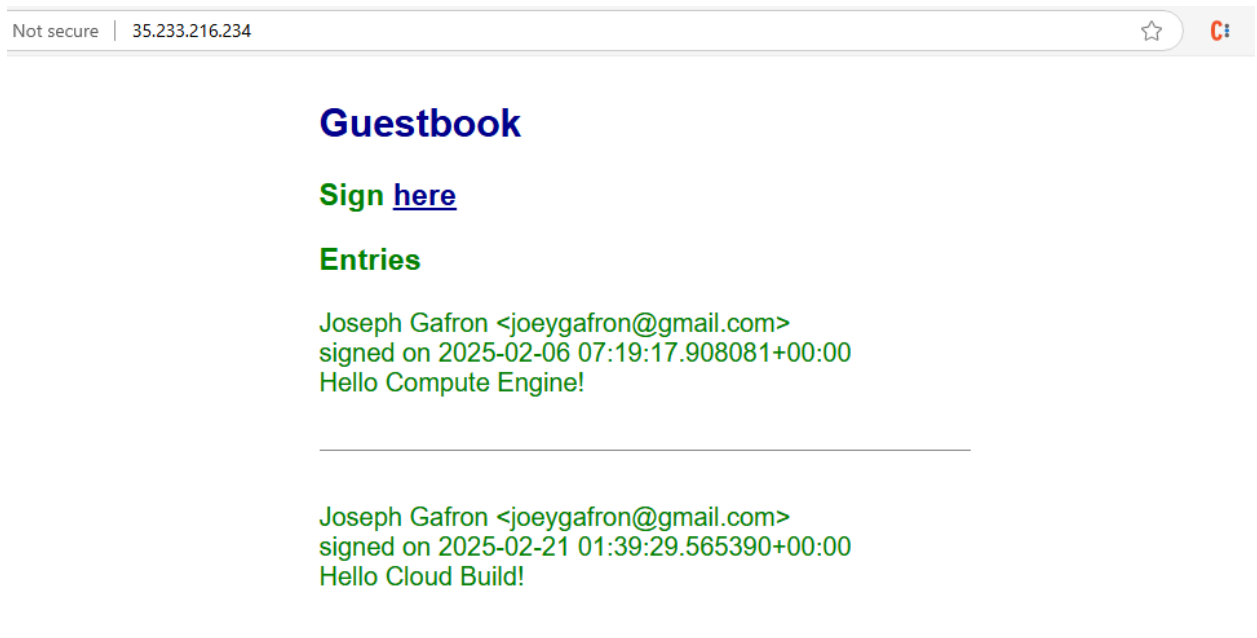
VM instance [gke-guestbook-default-pool-8b4f74bc-fth2](#) (Zone us-west1-b)

VM instance [gke-guestbook-default-pool-8b4f74bc-fth2](#) (Zone us-west1-b)

This is the one associated with the load-balancer:

Forwarding rule
[a40cff9d961224d4896f631e5500674d](#)

- Take a screenshot of the Guestbook including the URL with the entry in it.



- Does Google provide a Python package specifically for accessing the Knowledge Graph API?

No

- Show the source line that constructs the query we wish to send to the Knowledge Graph API.

```
req = kgsearch.entities().search(query=query, limit=1)
```

- Show the source line that then executes the query and saves the response. What is the name of the method that sends the query to the Knowledge Graph API?

```
res = req.execute()
```

```
return format_slack_message(query, res)
```

- What is the Python data type that is used to represent the formatted message?

A dictionary

- What are the three main attributes of the formatted message passed back to Slack?

Response_type, text, attachments

- Take a screenshot of its response for your lab notebook.



- Show the output for your lab notebook

```
(env) jgafron@cloudshell:~/python-docs-samples/vision/snippets/detect (cloud-gafron-jgafron)$ python detect.py label
Labels:
Bird
Ratite
Common ostrich
Flightless bird
Beak
Vertebrate
Wildlife
Terrestrial animal
Greater rhea
Feather
(env) jgafron@cloudshell:~/python-docs-samples/vision/snippets/detect (cloud-gafron-jgafron)$
```

- What is the name of the function?

detect_labels

- What type of Vision client is instantiated in it?

vision.ImageAnnotatorClient()

- What method is invoked in the Vision client to perform the detection?

label_detection

- What is the name of the attribute in the response object that contains the results we seek?

Label_annotations

- Take a screenshot of the output for the above commands

```
(env) jgafron@cloudshell:~/python-docs-samples/vision/snippets/detect
-university-logo-hd-png-download-removebg-preview-300x244.png
Logos:
Yale University
Yale University
Yale University
```

- What method is invoked in the Vision client to perform the detection?

Detect_logos

- Show the output for your lab notebook

```
(env) jgafron@cloudshell:~/python-docs-samples/speech/snippets (cloud-gafron-jgafron)$ python transcribe.py resources/audio.raw
Transcript: how old is the Brooklyn Bridge
```

- What is the name of the function?

transcribe_file

- What method is invoked in the Speech client to perform the detection?

client.recognize

- What is the name of the attribute in the response object that contains the results we seek?

Result.alternatives[0]

- Show the output for your lab notebook

```
(env) jgafron@cloudshell:~/python-docs-samples/translate/samples/snippets (cloud-gafron-jgafron)$ python snippets.py
Text: 你有沒有帶外套
Translation: Do you have a coat?
Detected source language: zh-TW
```

- What is the name of the function?

detect_language

- What method is invoked in the Translate client to perform the detection?

detect_language

- What is the name of the attribute in the response object that contains the results we seek?

result("Language")

- Show the output for your lab notebook

```
python language.py 'the protestors in Oregon put on gas masks and wore yellow t-shirts'
"homework is awful!" has sentiment=-0.800000011920929

Entities are:
name: homework
"homework is ok" has sentiment=0.30000001192092896

Entities are:
name: homework
"homework is awesome?" has sentiment=0.4000000059604645

Entities are:
name: homework
"homework is awesome!" has sentiment=0.8999999761581421

Entities are:
name: homework
"The protestors in Oregon put on gas masks and wore yellow t-shirts" has sentiment=-0.6000000238418579

Entities are:
name: protestors
name: gas masks
name: Oregon
name: t-shirts
(venv) isafron@cloudshell:~/python-deeps-samples/translate/samples/snippets/(cloud-isafron-isafron)$
```

- What is the name of the function that performs the transcription?

transcribe_gcs

- What is the name of the function that performs the translation?

translate_text

- What is the name of the function that performs the entity analysis on the translation?

entities_text

- What is the name of the function that performs the entity analysis on the image?

Comapare_audio_to_image

- If the program deems them unrelated, then based on the results from the APIs, what must be changed in the program to address this?

We need to add some sort of module or library to that will use word embeddings to recognize related words

- If the program deems them unrelated, then based on the results from the APIs, what must be changed in the program to address this?

We need to add a module or library that is able to use synonym matching to realize that "bike" and "bicycle" are the same thing

- **If the program deems them unrelated, then based on the results from the APIs, what must be changed in the program to address this?**

We need to add a module or library that allows for partial matching (the entity analysis returned "ostriches" instead of "ostrich").

- **What are the 3 labels with the highest confidence that the Video Intelligence API associates with the video and what are the confidences for each?**

Player

Audience

Games

- **What is the name of the client class in the package that is used?**

vidoeintelligence.VideoIntelligenceServiceClient()

- **What method is used in that class to perform the annotation?**

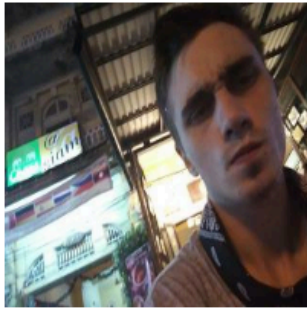
.annotate_video

- **Take a screenshot for your lab notebook that includes the URL.**

Google Cloud Platform - Face Detection Sample

This Python Flask application demonstrates App Engine Flexible, Google Cloud Storage, Datastore, and the Cloud Vision API.

Upload File: No file chosen



1556931889724.jpg was uploaded 2025-02-21 03:24:09.570306+00:00.

Joy Likelihood for Face: Very Unlikely

- **What line of code creates the query for previous detections?**

```
Query = datastore.client.query(kind="faces")
```

- **What line of code sends the query to Cloud Datastore?**

```
Image_entities = list(query.fetch())
```

Then, view the `upload_photo` route.

- **Show the line that retrieves the name of the storage bucket to use.**

```
CLOUD_STORAGE_BUCKET = os.environ.get("CLOUD_STORAGE_BUCKET")
```

- **What form field is used to specify the uploaded photo?**

```
Photo = request.files["file"]
```

- **Show the line that copies the photo's contents to the storage bucket.**

```
blob.upload_from_string(photo.read(), content_type=photo.content_type)
```

- **What method in Vision's annotation client is used to perform the analysis?**

Faces = vision_client.face_detection(image=image).face_annotations

- **What fields are stored in Cloud Datastore for each image?**

"blob_name,image_public_url,timestamp,joy"

- **What happens at the end of the `upload_photo` route?**

The program stores the image in the dataastore, then redirects the user to the homepage